

DNS Management and Name Resolution Configuration Lab

Lab Overview

This hands-on Windows Server 2022 lab focused on configuring and managing DNS services within an Active Directory environment. The lab covered Host (A) record creation, Reverse Lookup Zone configuration, PTR record creation, and DNS name resolution testing using command-line tools.

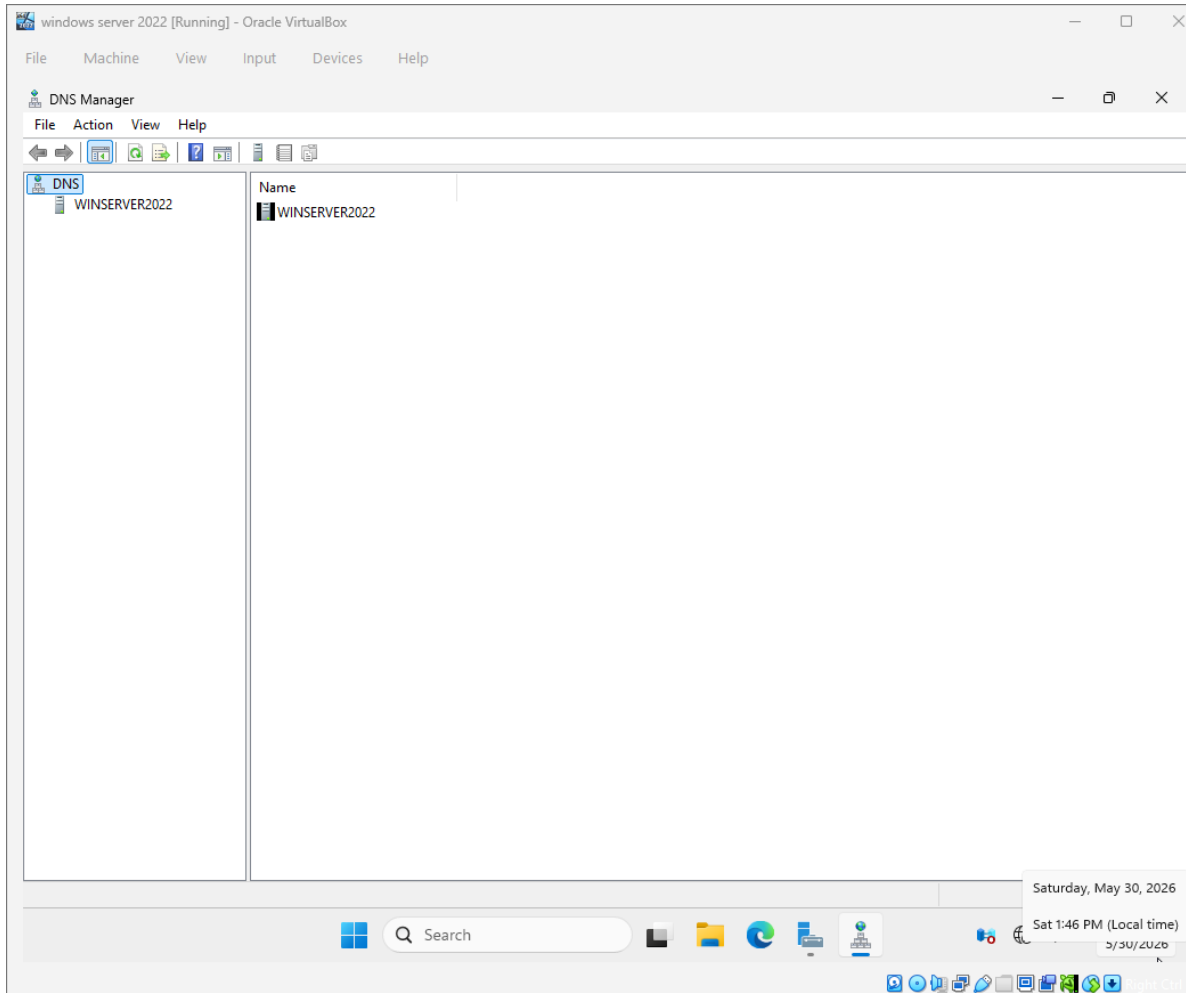
Topics Covered

- DNS Manager Administration
- Forward Lookup Zone Management
- DNS Host (A) Record Creation
- Fileserver Host Record Configuration
- DNS Resolution Testing
- NSLookup Testing
- Ping Testing
- Reverse Lookup Zone Configuration
- PTR Record Creation
- Reverse DNS Resolution
- Active Directory Integrated DNS
- DNS Troubleshooting and Verification

Conclusion

This lab successfully demonstrated DNS administration in a Windows Server 2022 environment. Practical experience was gained in managing DNS records, configuring forward and reverse lookup zones, implementing PTR records, and validating DNS functionality through testing and troubleshooting.

Screen shots



windows server 2022 [Running] - Oracle VirtualBox

FileMachineViewInputDevicesHelp

DNS Manager

FileActionViewHelp

DNS

WINSERVER2022

> Forward Lookup Zones

> Reverse Lookup Zones

> 1.168.192.in-addr.arpa

> Trust Points

> Conditional Forwarders

Name	Type	Data	Timestamp
(same as parent folder)	Start of Authority (SOA)	[1], winserver2022.sudeep....	static
(same as parent folder)	Name Server (NS)	winserver2022.sudeep.local	static

New Resource Record

Pointer (PTR)

Host IP Address:
192.168.1.

Fully qualified domain name (FQDN):
1.168.192.in-addr.arpa

Host name:

Browse...

☐

Allow any authenticated user to update all DNS records with the same name. This setting applies only to DNS records for a new name.

OK

Cancel

windows server 2022 [Running] - Oracle VirtualBox

FileMachineViewInputDevicesHelp

DNS Manager

FileActionViewHelp

DNS

WINSERVER2022

Forward Lookup Zones

Reverse Lookup Zones

1.168.192.in-addr.arpa

Trust Points

Conditional Forwarders

Name	Type	Data	Timestamp
(same as parent folder)	Start of Authority (SOA)	[1], winserver2022.sudeep....	static
(same as parent folder)	Name Server (NS)	winserver2022.sudeep.local.	static
192.168.1.10	Pointer (PTR)	fileserver.sudeep.local	

DNS Manager

File

Administrator: C:\WINDOWS\ X + v

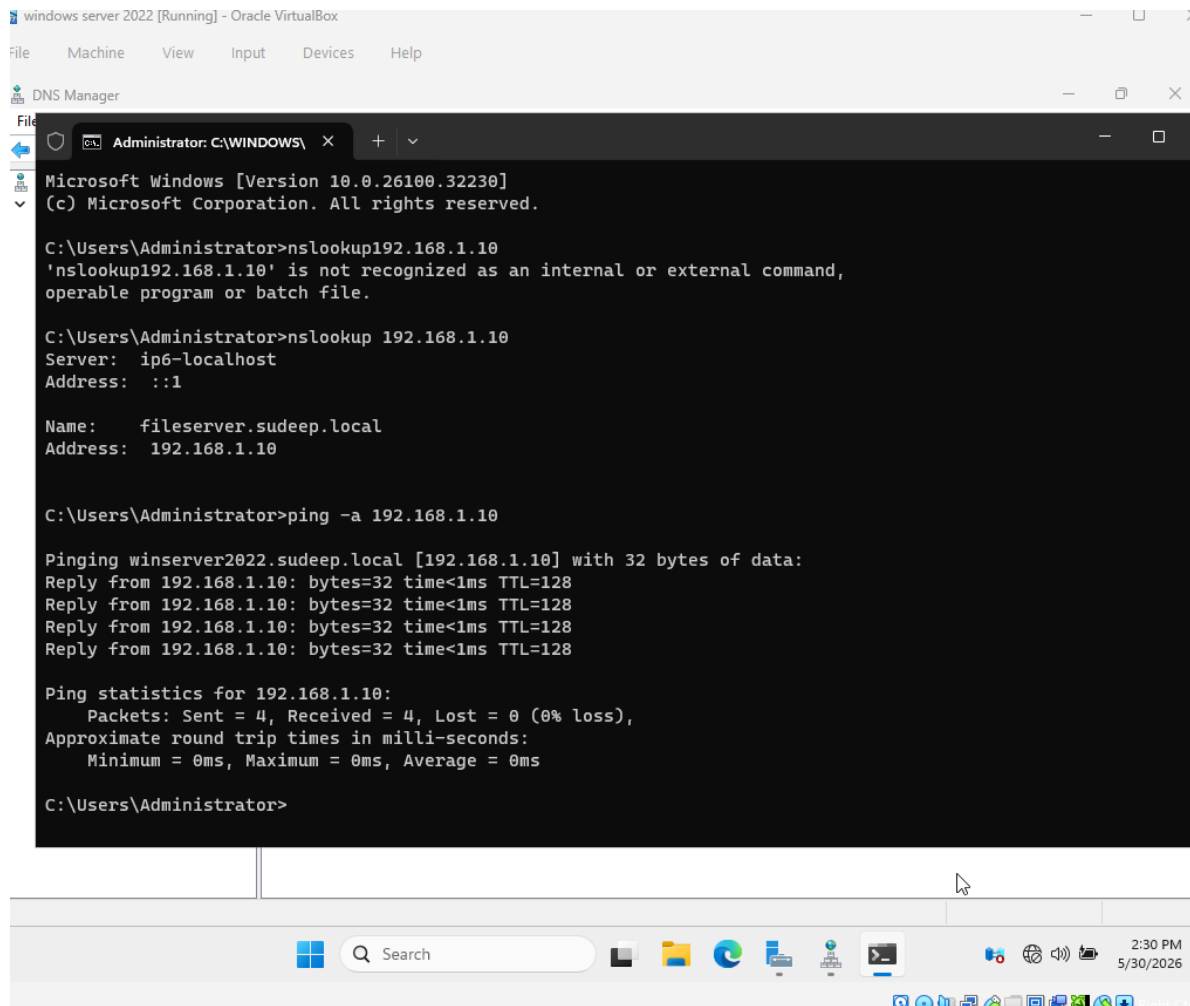
Microsoft Windows [Version 10.0.26100.32230]
(c) Microsoft Corporation. All rights reserved.

```
C:\Users\Administrator>nslookup192.168.1.10
'nslookup192.168.1.10' is not recognized as an internal or external
operable program or batch file.

C:\Users\Administrator>nslookup 192.168.1.10
Server:  ip6-localhost
Address:  ::1

Name:    fileserver.sudeep.local
Address: 192.168.1.10

C:\Users\Administrator>
```



MachineViewInputDevicesHelp

DNS Manager

FileActionViewHelp

DNS

WINSERVER2022

Forward Lookup Zones

Reverse Lookup Zones

1.168.192.in-addr.arpa

Trust Points

Conditional Forwarders

Name	Type	Data	Timestamp
(same as parent folder)	Start of Authority (SOA)	[1], winserver2022.sudeep....	static
(same as parent folder)	Name Server (NS)	winserver2022.sudeep.local	static

New Resource Record

Pointer (PTR)

Host IP Address:
192.168.1.10

Fully qualified domain name (FQDN):
10.1.168.192.in-addr.arpa

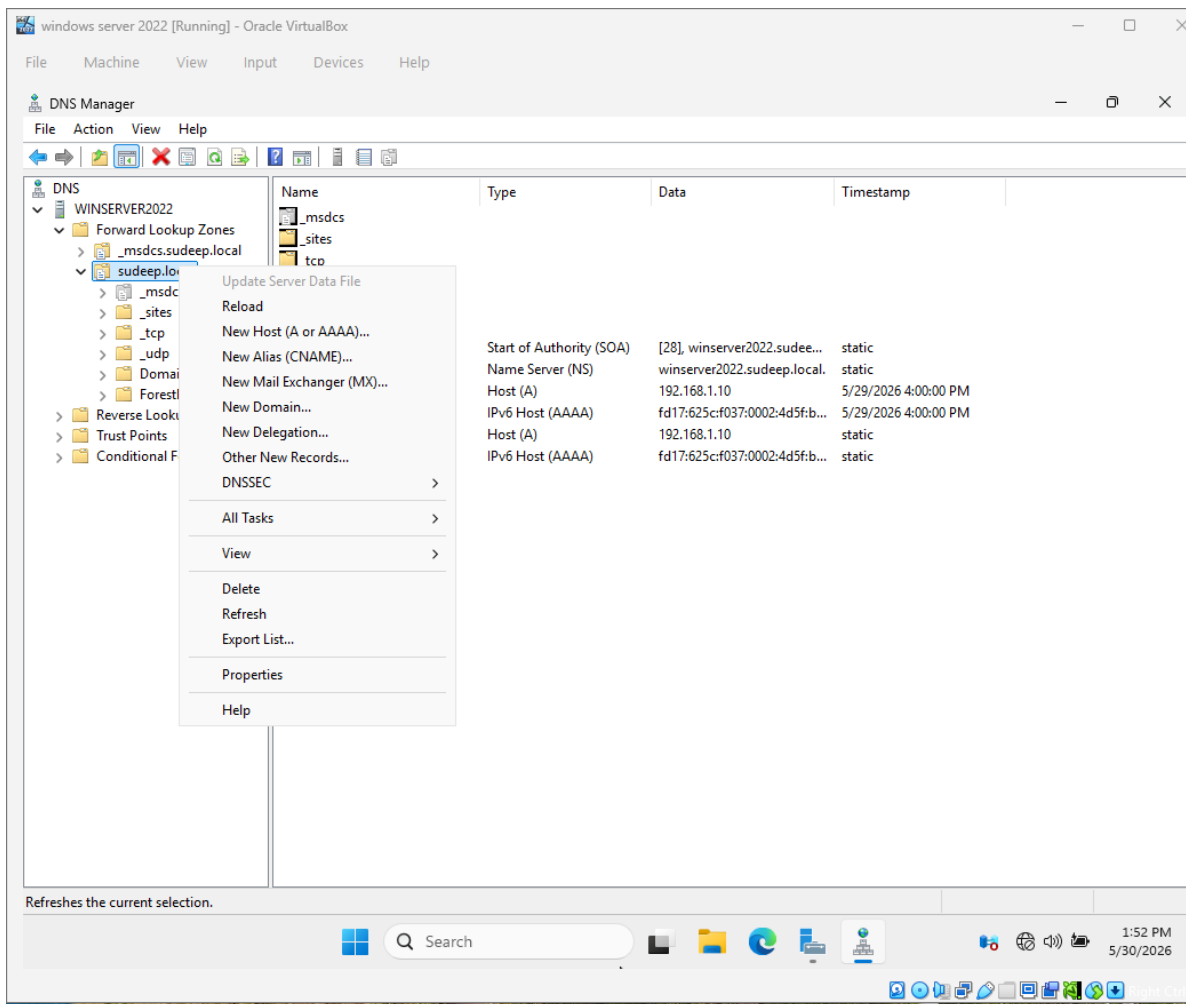
Host name:
fileserver.sudeep.local

Browse...

☐ Allow any authenticated user to update all DNS records with the same name. This setting applies only to DNS records for a new name.

OKCancel

Speakers (High



Windows Server 2022 [running] - Oracle VM VirtualBox

File Machine View Input Devices Help

DNS Manager

File Action View Help

DNS

- WINSERVER2022
 - Forward Lookup Zones
 - _msdcs.sudeep.local
 - sudeep.local
 - _msdcs
 - _sites
 - _tcp
 - _udp
 - DomainDnsZones
 - ForestDnsZones
 - Reverse Lookup Zones
 - Trust Points
 - Conditional Forwarders

Name	Type	Data	Timestamp
_msdcs			
_sites			
_tcp			
_udp			
DomainDnsZones			
ForestDnsZones			
(same as parent folder)	Start of Authority (SOA)	[28], winserver2022.sudee...	static
(same as parent folder)	Name Server (NS)	winserver2022.sudeep.local.	static
(same as parent folder)	Host (A)	192.168.1.10	5/29/2026 4:00:00 PM
(same as parent folder)	IPv6 Host (AAAA)	fd17:625c:f037:0002:4d5f:b...	5/29/2026 4:00:00 PM
winserver2022	Host (A)	192.168.1.10	static
winserver2022	IPv6 Host (AAAA)	fd17:625c:f037:0002:4d5f:b...	static
fileserver	Host (A)	192.168.1.10	

DNS Manager

1:56 PM 5/30/2026

